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GIS Portfolio 2026

Map 1: Intro to GIS, Fall 2023 final project



This map was created by integrating multiple datasets, including Washington State major road networks, topographic data, and a custom dataset of Costco warehouse locations that I compiled. I gathered approximately 30 warehouse addresses across Washington State from the official Costco website and geocoded them into latitude and longitude coordinates using Google Earth. The resulting data was organized into a CSV file for spatial analysis and mapping.

The map itself was produced in QGIS, where I combined and visualized the datasets. Final design elements, including the addition of the Costco logo and visual refinements, were completed in Adobe Photoshop.

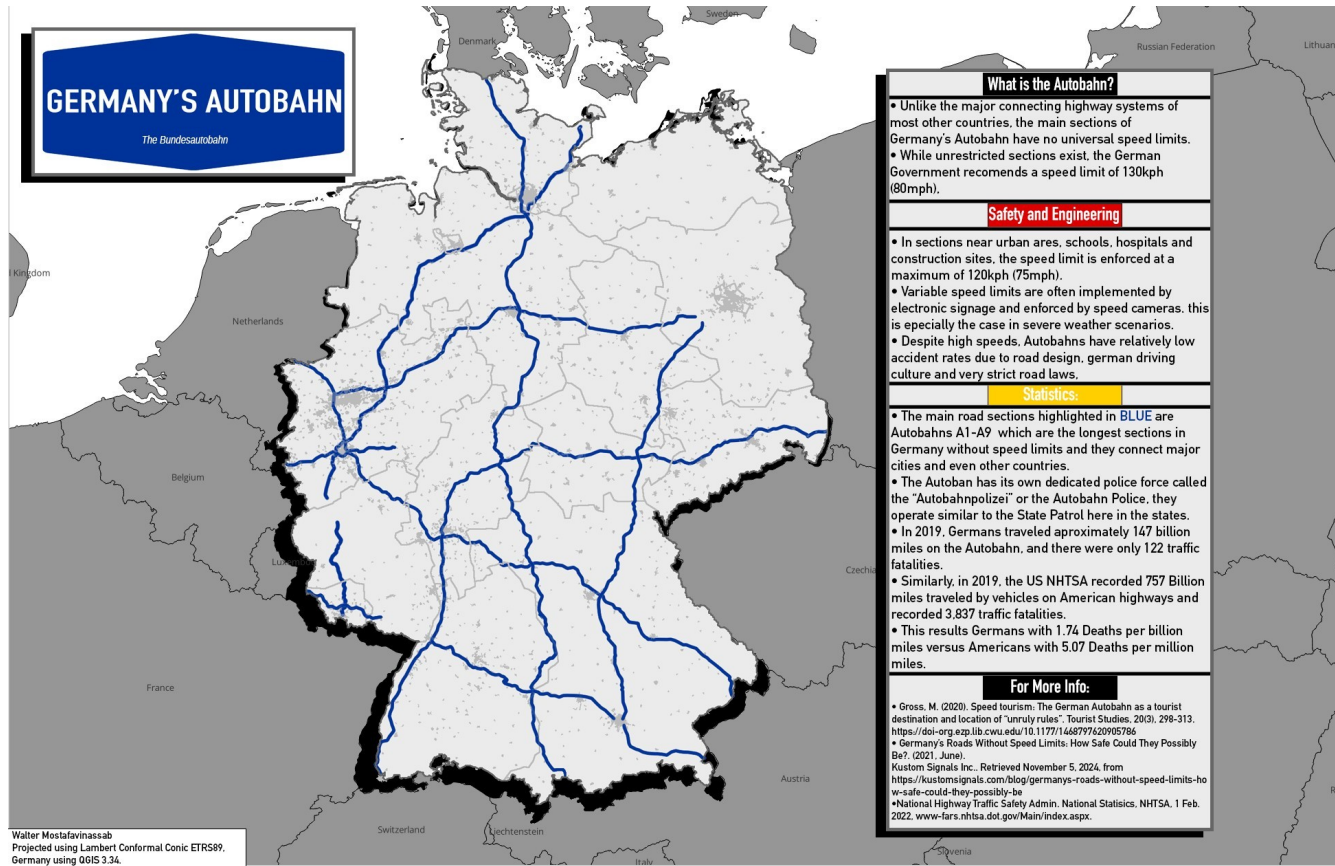
This project highlights my ability to collect and process spatial data, perform geocoding, and create clear, visually engaging maps using GIS and design tools.

Custom Costco Dataset CSV:

	A	B	C	D	E	F	G	H	I
1	Latitude	Longitude	City Name	Address	Zip Code	Phone#	Onion Dispensers	Onion Cups	Warehouse Franchise #
2	48.7922024	-122.51466	Bellingham	4125 ARCTIC AVE	98226 (360) 671-6947		0	0	1216
3	47.16720265	-122.1587204	Bonney Lake	9801 204TH AVE E	98391 (253) 750-8120		0	1	1278
4	48.45407401	-122.3373144	Burlington	1725 S BURLINGTON BLVD	98233 (360) 757-5703		0	1	662
5	46.42302454	-117.0429413	Clarkston	301 5TH ST	99403 (509) 758-1800		0	1	103
6	47.35547881	-122.1220598	Covington	27520 COVINGTON WAY SE	98042 (253) 796-1004		0	1	1029
7	45.62257073	-122.4718861	Camas	19610 SE 1ST ST	98607 (360) 258-6221		0	1	1086
8	47.90407631	-122.2097052	Everett	10200 19TH AVE SE	98208 (425) 379-7451		0	1	114
9	47.28693078	-122.3100519	Federal Way	35100 ENCHANTED PKWY S	98003 (253) 874-3652		0	1	61
10	47.23822152	-122.3733029	Fife (Buisness center)	3900 20TH ST E	98424 (253) 719-1950	N/A	N/A		767
11	47.35798803	-122.6038771	Gig Harbor	10990 HARBOR HILL DR NW	98332 (253) 853-8600		0	1	624
12	47.55128809	-122.0526699	Issaquah	1801 10TH AVE NW	98027 (425) 313-0965		0	1	110
13	46.22551792	-119.2350464	Kennewick	8505 W. GAGE BLVD	99336 (509) 737-8861	?	?		486
14	47.68077182	-122.1816556	Kirkland	8629 120TH AVE NE	98033 (425) 827-1693		0	1	8
15	47.05996251	-122.760178	Lacey	1470 MARVIN RD NE	98516 (360) 412-3500		0	1	740
16	47.97519555	-122.1061637	Lake Stevens	9210 24TH STREET SE	98258 (425) 367-6422		0	1	1588
17	47.83416983	-122.2730498	Lynnwood	18109 33RD AVE W	98037 (425) 697-1062		0	1	1190
18	47.82525461	-122.3105987	Lynnwood (Buisness center)	19105 HIGHWAY 99	98036 (425) 640-7700	N/A	N/A		115
19	48.1475464	-122.1917006	Marysville	16616 TWIN LAKES AVE	98271 (360) 652-4532		0	1	642
20	47.15575268	-122.3078253	Puyallup	1201 39TH AVE SW	98373 (253) 445-7543		0	1	660
21	47.67348862	-122.0910862	Redmond	7725 188TH AVE NE	98052 (425) 406-5382		0	1	1225
22	47.56537205	-122.3303746	Seattle	4401 4TH AVE S	98134 (206) 622-3136		0	1	1
23	48.07710501	-123.1261304	Sequim	955 W WASHINGTON ST	98382 (360) 406-2025		0	1	639
24	47.77504828	-122.3452124	Shoreline	1175 N 205TH ST	98133 (206) 546-0480		1	1	106
25	47.65666822	-122.6809084	Silverdale	10000 MICKELBERRY RD NW	98383 (360) 692-9213		0	1	13
26	47.65866196	-117.3336504	Spokane	5601 E SPRAGUE AVE	99212 (509) 842-0003		0	1	670
27	47.76722172	-117.3835294	Spokane	12020 N NEWPORT HWY	99218 (509) 466-9668		0	1	1298
28	47.22533532	-122.4680396	Tacoma	2219 S 37TH ST	98409 (253) 475-5595		0	1	95
29	47.44560557	-122.2487928	Tukwila	400 COSTCO DR STE 150	98188 (206) 575-9191		0	1	6
30	46.99672837	-122.916303	Tumwater	5500 LITTLEROCK RD SW	98512 (360) 357-6580		0	1	64
31	46.5621481	-120.5005457	Union Gap	2310 LONGFIBRE RD	98903 (509) 454-5242		0	1	1013
32	45.68523686	-122.6034882	Vancouver	6720 NE 84TH ST	98665 (360) 828-2262		0	1	772
33	47.40139167	-120.2796889	Wenatchee	375 HIGHLINE DR	98802 (509) 886-0593		0	1	112
34	47.77955882	-122.1479531	Woodinville	24008 SNOHOMISH WOODINVILLE RD	98072 (425) 806-7700		0	1	747

This attribute table displays the processed dataset of Costco warehouse locations across Washington State. Each record includes the geocoded latitude and longitude coordinates derived from address data, along with associated site identifiers. The table was used to structure the point dataset for spatial analysis and visualization in QGIS.

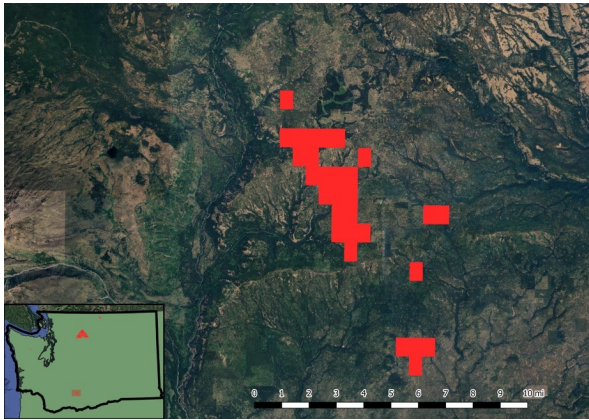
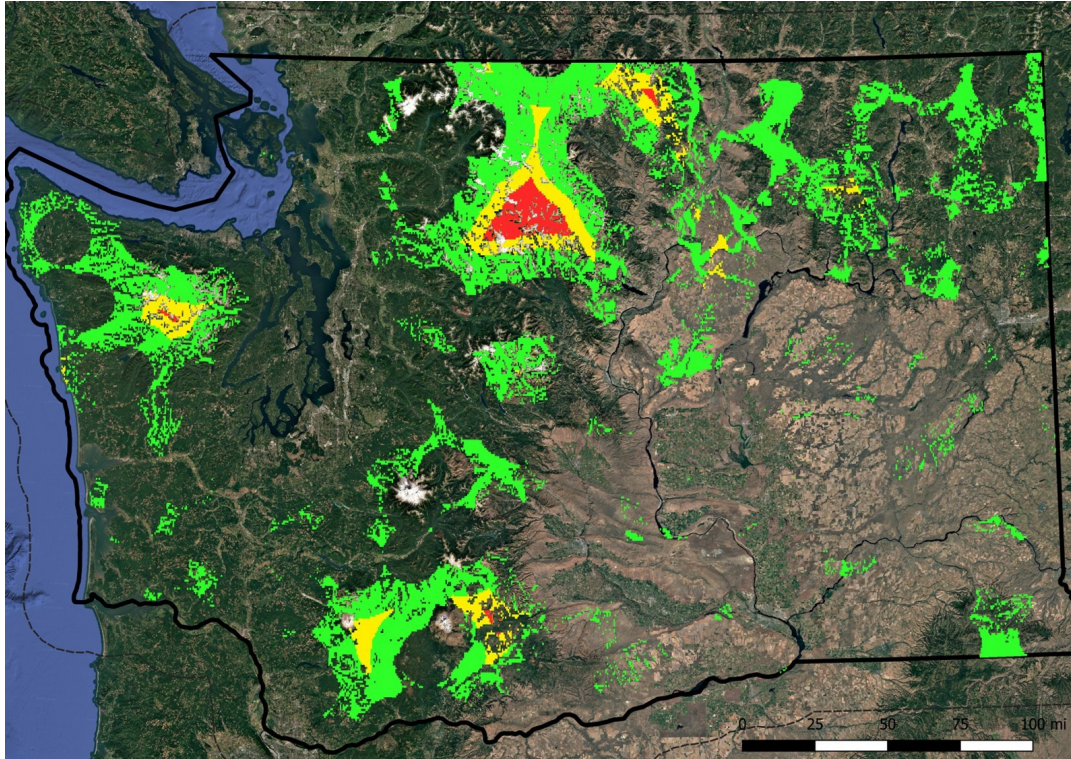
Map 2: Spring 2024: Transportation Geography final project. Autobahn mapping.



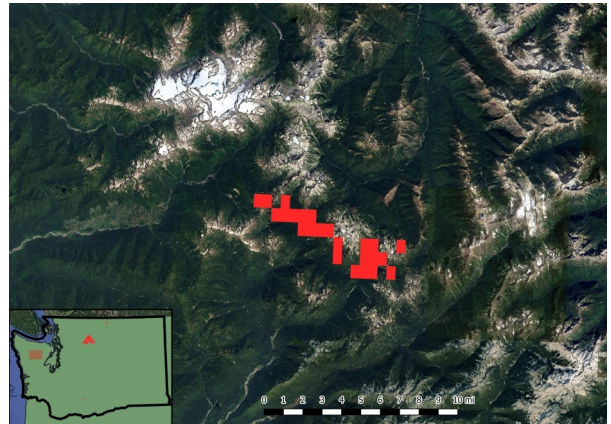
This project involved designing a transportation-focused infographic using geospatial data. I combined a global basemap with a dataset of major road networks in Germany to highlight the country's transportation infrastructure within a broader geographic context. I also wanted to challenge myself by doing something that was not bound within the United States. The map was created using QGIS for data processing, spatial analysis, and layout design, while Adobe Photoshop 2022 was used for the visual presentation.

This project demonstrates my ability to integrate multiple datasets, apply cartographic design principles, and produce visually engaging, informative graphics.

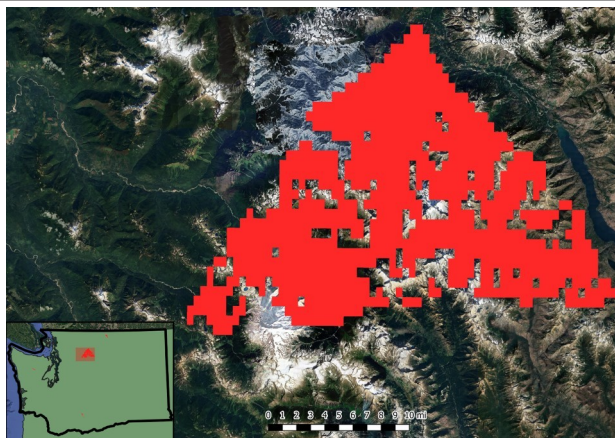
Map 3: Spring 2025 Final Project: Cellular network no-coverage mapping



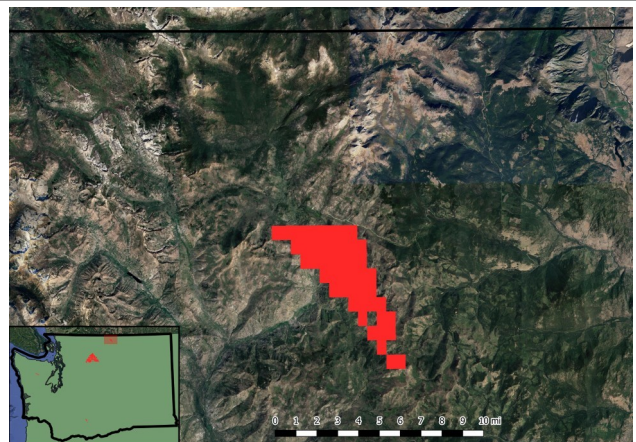
Site 1: Around Mount Adams



Site 2: East of Mount Olympus



Site 3: Glacier Peak Wilderness



Site 4: Tonasket/ Okanogan County

This is the project I am most proud of, as it combines spatial analysis, data processing, and real-world problem solving. The goal was to identify areas in Washington State with limited or no cellular coverage by modeling line-of-sight signal availability using viewshed analysis.

To accomplish this, I used a Digital Elevation Model (DEM) alongside two large datasets of cell tower locations in both the USA and Canada downloaded from the FCC website and the CRTC website respectively. The analysis involved generating individual viewshed rasters for each tower, representing the areas theoretically visible within a 20-mile radius and then aggregating over 800 of these rasters into a single statewide coverage map using Python.

The data preparation process was equally important. I began with a nationwide dataset of over 20,000 cell towers of the USA and Canada and clipped it to Washington State. To account for signal overlap from neighboring regions, I created a 30-mile buffer around the state boundary, ensuring that towers in adjacent areas of Oregon and Idaho were included. After applying this constraint, the working dataset was reduced to approximately 800 relevant towers.

The final output was a 3 color classified coverage map highlighting potential service gaps: green areas indicating possible coverage gaps, yellow areas suggesting likely gaps, and red areas representing high-confidence no-coverage zones. Focusing on the most critical results, I identified four distinct locations categorized as definite no-coverage areas.

For tools and workflow, I used ArcPy for automation and analysis, QGIS and ArcGIS Pro for cartographic rendering, and Microsoft PowerPoint to present my findings.

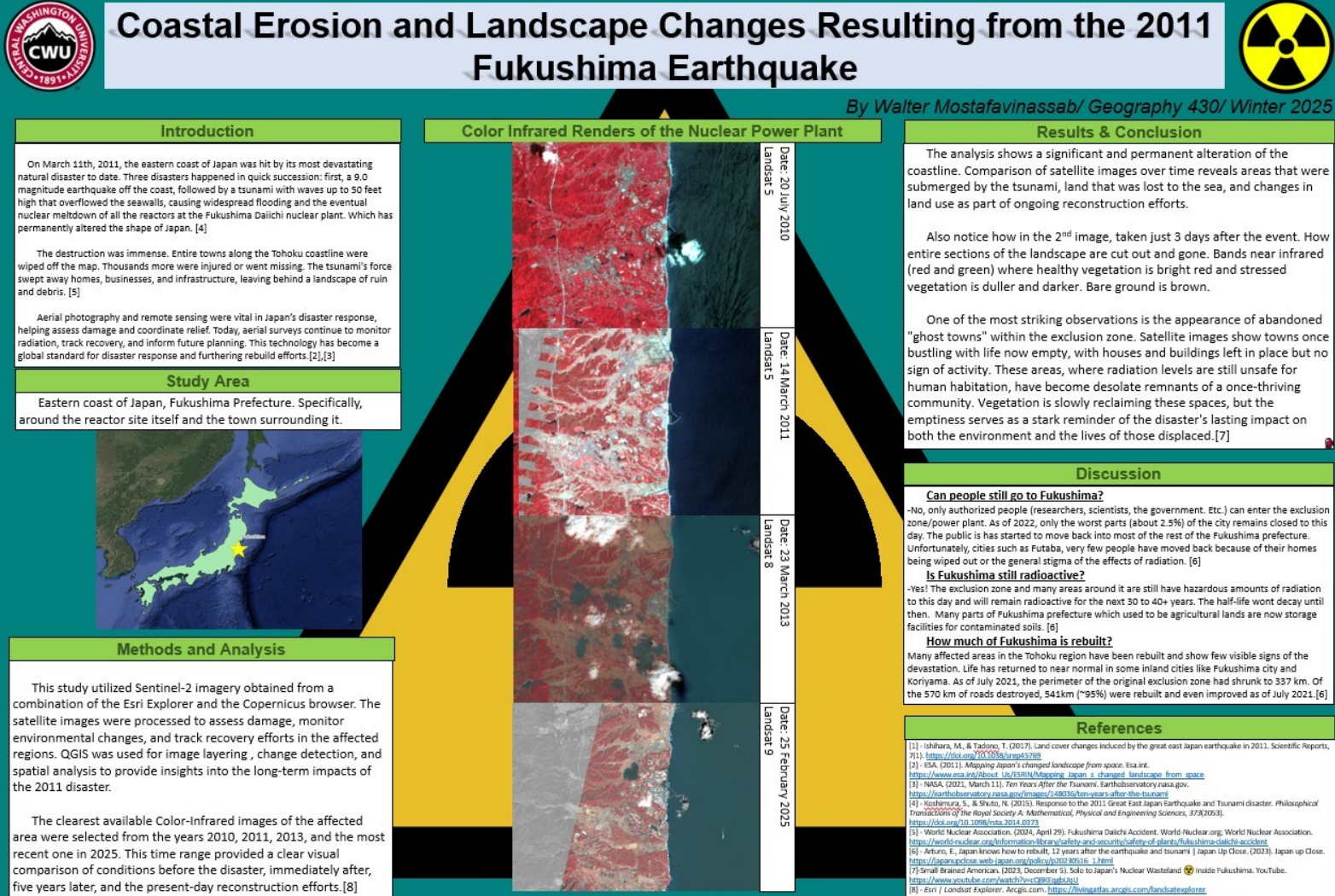
This project demonstrates my ability to handle large geospatial datasets, automate complex analyses, and communicate results effectively through both maps and presentations.

Python Code used for the viewshed automated batch processing:

```
batch_viewshed.py × batch_tif_combiner.py
1 import arcpy
2 from arcpy.sa import *
3
4 # Set environment and inputs
5 arcpy.env.workspace = r"C:\Users\Hans_Gloopa\Documents\celltowers"
6 arcpy.env.overwriteOutput = True
7
8 dem = r"C:\Users\Real_Slim_Shady\Documents\celltowers\usaDEM.tif"
9 towers = r"C:\Users\Real_Slim_Shady\Documents\celltowers\CANUSTowers.shp"
10 output_folder = r"C:\Users\Real_Slim_Shady\Documents\celltowers\output"
11
12 # Check out Spatial Analyst extension
13 arcpy.CheckOutExtension("Spatial")
14
15 # Loop through each tower point using SearchCursor
16 with arcpy.da.SearchCursor(towers, field_names: ['FID', 'SHAPE@XY']) as cursor:
17     for row in cursor:
18         tower_id = row[0]
19         tower_xy = row[1]
20         print(f"Processing viewshed for tower ID {tower_id} at location {tower_xy}")
21
22         # Create a feature layer for the single tower by selecting where FID = tower_id
23         where_clause = f"FID = {tower_id}"
24         temp_layer_name = f"tower_{tower_id}"
25         arcpy.management.MakeFeatureLayer(towers, temp_layer_name, where_clause)
26
27         # Run the viewshed tool
28         out_raster_path = f"{output_folder}\\viewshed_{tower_id}.tif"
29         viewshed_result = arcpy.sa.Viewshed(dem, temp_layer_name)
30         viewshed_result.save(out_raster_path)
31
32         # Delete temporary layer to clean up
33         arcpy.management.Delete(temp_layer_name)
34
35 # Check in Spatial Analyst extension
36 arcpy.CheckInExtension("Spatial")
37
38 print("Batch viewshed analysis complete.")
```

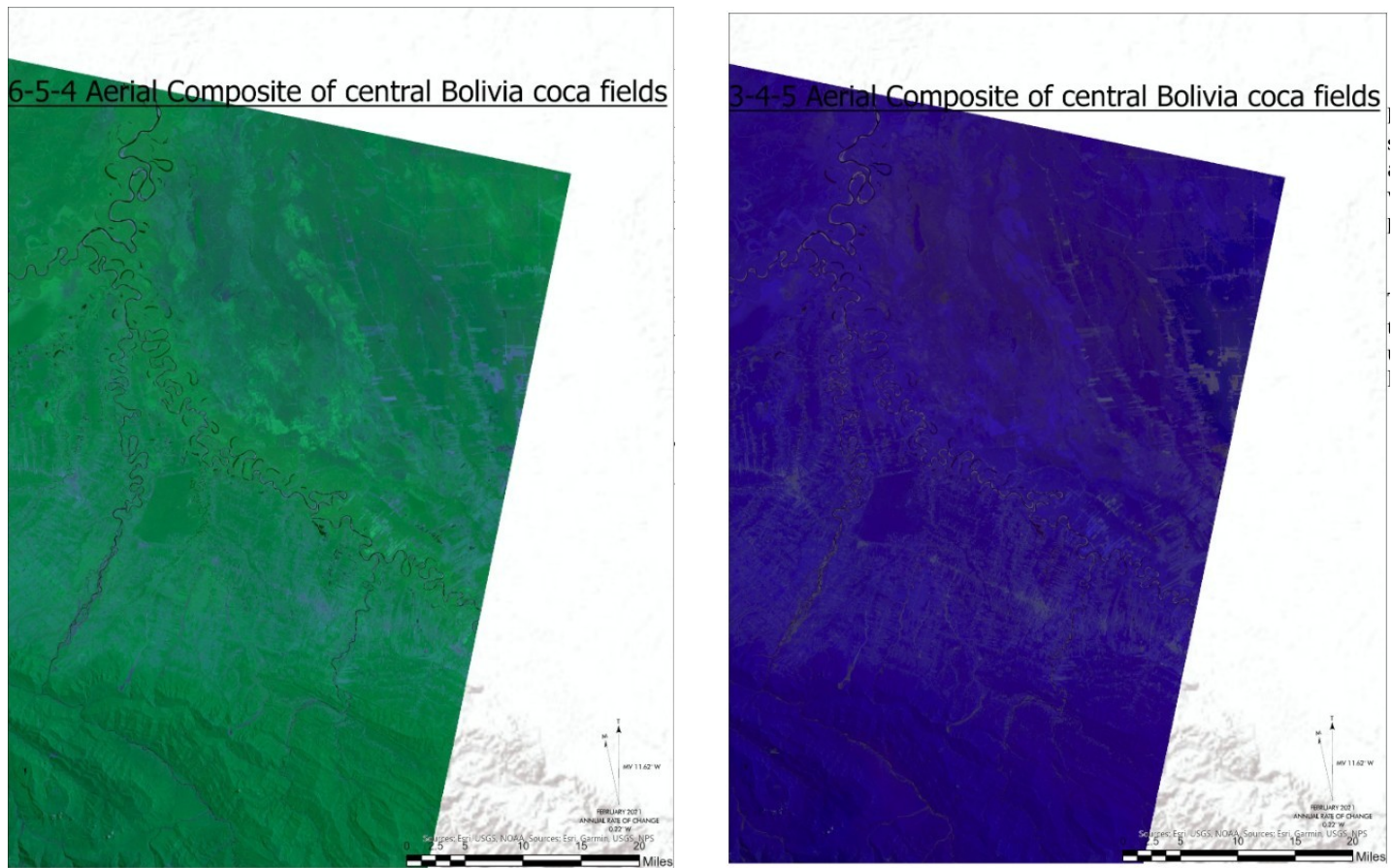
Wrote a python script to batch process GDAL viewshed for each tower. Then combined all of the viewshed rasters with get CellStatistics() in arcpy to make one big raster. Resulting in the cumulative line of sight visibility

Map 4: Winter 2025 Remote Sensing final project



This map I took color infrared images of the Fukushima Daiichi powerplant zone. This study utilized Sentinel-2 imagery obtained from a combination of the Esri Explorer and the Copernicus browser. The satellite images were processed to assess damage, monitor environmental changes, and track recovery efforts in the affected regions. QGIS was used for image layering, change detection, and spatial analysis to provide insights into the long-term impacts of the 2011 disaster. The clearest available Color-Infrared images of the affected area were selected from the years 2010, 2011, 2013, and the most recent one in 2025. This time range provided a clear visual comparison of conditions before the disaster, immediately after, 3 years later, and the present-day reconstruction efforts.

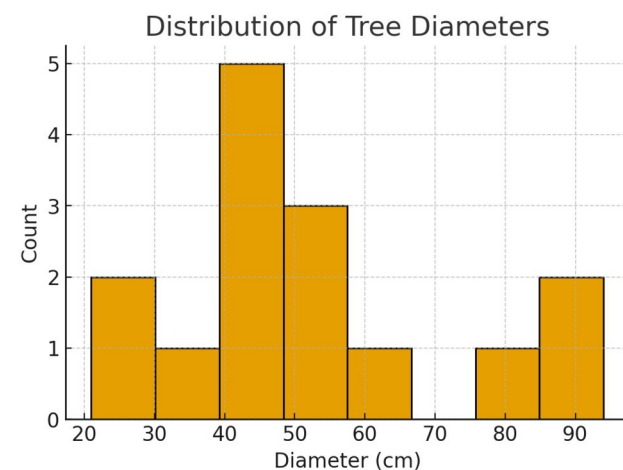
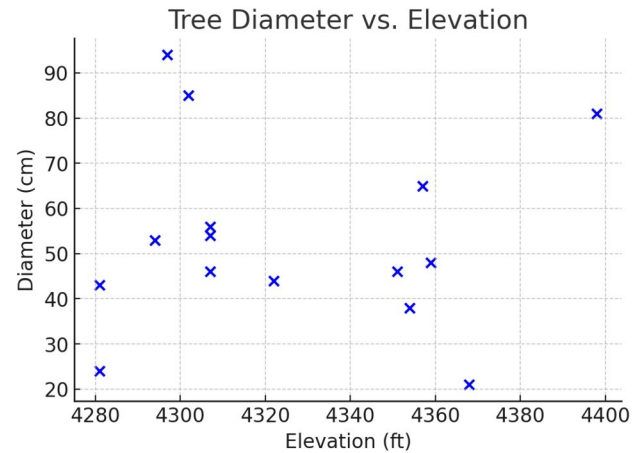
Map 5: Winter 2025 Finding areas of illicit coca production in Bolivia using different infrared bands.



This project analyzes potential illicit coca cultivation in central Bolivia using satellite imagery and remote sensing techniques. The study area is located in the Andes at approximately 17.3875° S, 64.5619° W. The analysis uses imagery from Landsat 9 collected on October 4, 2024, with 16-bit radiometric resolution and 11 spectral bands. Using ArcGIS Pro, I produced two maps to differentiate likely coca-growing areas from other agricultural land. Bands 4, 5, and 6 were used to highlight vegetation health and moisture through infrared reflectance, helping identify dense crop areas. Bands 3, 4, and 5 were used to emphasize vegetation patterns, allowing comparison between organized agricultural fields and more irregular plots associated with illicit cultivation. The results indicate that suspected coca fields appear as smaller, less structured patches, primarily in the northeastern portion of the study area, while larger and more uniform fields north of the river suggest formal agriculture.

This project highlights my ability to analyze multispectral imagery, apply remote sensing methods, and interpret land-use patterns using GIS tools.

Map 6: Fall 2025 Ponderosa Pine Field Survey



This study investigated whether ponderosa pine (*Pinus ponderosa*) trees at lower elevations show greater growth, used as a proxy for nutrient availability, compared to those at higher elevations within the same forest. A total of 15 trees were measured along a 100-meter transect spanning an elevation gradient of approximately 100 feet. For each tree, diameter at breast height (DBH), height, and spatial position were recorded. Although all 15 trees were recorded in the dataset, a subset of three representative trees located at the top, middle, and bottom of the slope were selected for core sampling. Increment cores were extracted at breast height to analyze growth ring patterns. Each core was labeled by elevation position and stored for later analysis. This stratified transect design enabled direct comparison of growth across the elevation gradient using consistent sampling methods.

Tree locations were recorded using a personal Garmin GPSMAP 66sr and later compiled in QGIS to produce a spatial dataset of the study area. Field measurements and tree diameter data were then organized and analyzed in Microsoft Excel to generate graphs and visualize trends in tree growth across elevation.

Writing Sample: Spring 2025 Pallas's Cat Biogeography Paper

Producing this paper required a combination of research, analytical, and technical skills that are highly relevant to environmental and geospatial science roles. The project involved synthesizing peer-reviewed ecological literature, interpreting species distribution and evolutionary data, and communicating complex biogeographic relationships in a clear, structured format. It also required strong scientific writing skills to organize information across ecology, evolution, and conservation while maintaining accurate citation practices. From a technical perspective, the work involved interpreting spatial range maps and ecological datasets, supporting spatial reasoning about habitat distribution, elevation, and ecosystem interactions. In addition, the project required data literacy and critical thinking to evaluate sources, identify key environmental threats, and connect human impacts to ecological outcomes. Probably my favorite paper because I got to talk all about my favorite animal.

This assignment demonstrates my ability to integrate scientific research, spatial reasoning, and clear communication to produce a cohesive analysis.



Species: *Otocolobus manul*

Common name: “Pallas’s Cat”

Section 1: Biogeographic overview.

Otocolobus manul, commonly name the Pallas’s cat or in many places simply the “manul,” is a small wild feline with thick grey fur, endemic to Central Asia, Mongolia, Siberia, Iran, and parts of the Middle East. Its fur color varies from grey to orange, depending on how far south the population resides (Pittet, 2024). This dense coat allows the cat to survive in cold, mountainous steppe climates and also provides camouflage in its rocky environment. The Pallas’s cat has a flat, striped face, small ears, and a compact build that helps it stay warm and blend into its surroundings (Figure 1). Pallas’s cats are considered active dispersers, moving independently across landscapes to find new territories especially during the juvenile stage. Kittens leave their mothers around 4–5 months old and travel significant distances to avoid resource competition and establish their own ranges. Some adults have been recorded migrating over 170 km in two months. Despite this mobility, Pallas’s cats are moderate to poor dispersers compared to other wild cats due to their strong habitat and dietary specialization. They rely on rocky steppes and prey like marmots and small rodents, making it hard to adapt to new areas. High kitten mortality (about 60% before dispersal) and threats such as predation and habitat loss further reduce their dispersal success. Litters typically consist of 3–4 kittens, though up to 8 have been recorded, and females can breed within a year. (Pittet 2024, Brown, 2005)

The Pallas’s cat is a unique member of the Felidae family, notable for its specialized adaptations to high-elevation, cold desert habitats. Its thick fur and short legs are well-suited to the cold, arid environments it inhabits. Pallas’s cats are native to Central Asia and have a wide but patchy distribution. Their range stretches from the Caucasus Mountains in the west to Mongolia and parts of western China. They are also found in parts of southern Russia, northern Iran, and throughout the

Tibetan Plateau (Ross et al., 2020). These cats are well adapted to high-altitude environments and can be found living at elevations up to 5,000 meters (about 16,400 feet). Although they cover a large geographic area, their populations are often scattered and isolated due to the fragmented nature of their habitat. The largest and most stable populations are found in Mongolia and parts of Russia, where the steppe ecosystems are more extensive and less disturbed by human activity. Pallas's cats are solitary and crepuscular meaning they are typically active at dawn and dusk and exhibit highly elusive behavior. Their primary prey includes pikas, small rodents, and birds, making them crucial predators in steppe ecosystems (Pittet, 2024). Despite its wide distribution (Figure 2), the manul is currently considered of Least Concern by the IUCN as of 2019. However, due to ongoing habitat degradation and human encroachment, the species was previously assessed as Near Threatened in 2016 (Ross et al 2020). Pallas's cats are known to migrate over long distances, primarily in search of areas with sufficient prey. These migrations are usually responses to prey depletion caused by cyclic fluctuations in rodent populations, particularly pikas, which are vulnerable to severe winters and summer droughts. Additionally, Pallas's cats may migrate to avoid predation, reduce competition with other carnivores, or colonize new territories (UN, 2024). The ecological niche of the Pallas's cat is defined by its adaptations to cold, arid environments in Central Asia, particularly steppes and rocky regions where pikas are abundant. Pallas's cats prefer cold, dry environments such as open steppe, rocky hillsides, and shrub-covered areas. They often live in places with plenty of cover, like rocky crevices or abandoned marmot burrows (Pittet, 2024), where they can hide from predators and harsh weather. Their reclusive nature helps them remain hidden from both predators and human disturbance (Ross, 2009). Mammals such as pikas, voles, and small bird species form a crucial part of the steppe food web and often undergo population booms and busts influenced by seasonal and climatic changes. The Pallas's cat is

well adapted to hunting in open terrain, using a stalk-and-ambush strategy that often involves concealment in rock crevices or burrows.

The Mongolian steppe itself hosts a rich and varied community of mammals. Large ungulates like Mongolian gazelles, red deer, and Argali sheep roam the grasslands, while the smaller mammal population includes species such as long-tailed sousliks, Mongolian gerbils, mountain voles, and Daurian pikas. Rodents and lagomorphs, including Brandt's vole, Pallas's pika, Siberian jerboas, and tolai hares, are particularly important as prey for mid-sized carnivores like the Pallas's cat. Larger species, like the Siberian marmot and Daurian hedgehog, also inhabit this ecosystem, though they may not be common targets for the Pallas's cat (Ross, 2009). Despite its efficiency as a hunter, the Pallas's cat faces threats from both aerial and terrestrial predators. Steppe eagles, in particular, pose a danger to juveniles and young adults, while red foxes and gray wolves may either prey on or compete with the cat for food. Other carnivores that share this habitat include the corsac fox, Eurasian lynx, Siberian and marbled polecats, and the Eurasian badger. The skies over the steppe are patrolled by an impressive array of raptors, including golden eagles, vultures, upland buzzards, eagle owls, and saker falcons. These birds contribute to the rich predator-prey dynamics of the region. (Ross, 2009) During spring and autumn migrations, wetlands like those along the Tuul River become vital foraging grounds for birds traveling to and from Lake Baikal, highlighting the seasonal shifts in biodiversity. Insect life also plays a role in this dynamic ecosystem. Grasshoppers and cicadas become especially abundant during hot, dry summers, while beetles round out the insect community, providing an additional food source for many species (Ross et al., 2012). Additionally, human-related threats such as poaching, habitat encroachment, and poisoning often aimed at rodents but affecting the manul indirectly have intensified in recent decades. Domestic dogs, especially in Mongolian agricultural communities, may chase or kill

Pallas's cats, further stressing their already fragmented populations. These interactions place the Pallas's cat in a delicate ecological balance, where both predators, humans and prey availability strongly influence its survival, movement patterns, and reproductive success. Understanding these relationships is key to conserving the species within its increasingly threatened steppe habitat (Ross et al 2012).

Section 2: History and Evolution

The Pallas's cat is the sole species in the genus *Otocolobus*. The species was first described by German naturalist Peter Simon Pallas in 1776, hence its common name. The cats have been historically understudied, largely due to their range lying within former Soviet territories, where environmental research was limited and data often inaccessible or unpublished. Only in the past 20–30 years have focused studies on the Pallas's cat increased, improving our understanding of its biogeography. There are no known fossils to this species. However, in recent history researchers have studied the evolutionary history of the Felidae family extensively by examining skull shapes and sizes across different cat species. Using statistical methods, they analyzed how skull shape correlates with skull size and overall body mass. The study revealed two major morphological patterns; large cats tend to have long, heavy snouts and massive skulls, while smaller cats typically have rounder heads and shorter faces (Sicuro 2011). Researchers identified these patterns by mapping specific anatomical landmarks on the skull and comparing how these points varied across species. Using the results of the study researchers created a phylogenetic tree for the felidae family. (figure 3) Recent advances in molecular phylogenetics have grouped the 33 modern species of wild cats into eight distinct evolutionary lineages (Johnson et al., 2006). These groupings have been largely supported by similarities in morphology, biology, and physiology. According to Johnson's 2006 analysis, the Pallas's cat belongs to the leopard

cat lineage, though it diverged early from the other members of this group, supporting its classification in a separate genus (Sicuro 2011). The Pallas's cat lineage is believed to have branched off from a common felid ancestor around 5.5 to 5.8 million years ago, while domestic cats diverged approximately 3.4 million years ago. Other species in the leopard cat lineage include the Asian leopard cat (*Prionailurus bengalensis*), the fishing cat (*Prionailurus viverrinus*), the flat-headed cat (*Prionailurus planiceps*), and the rusty-spotted cat (*Prionailurus rubiginosus*).

Without a fossil record specifically for the Pallas's cat, it's difficult to reconstruct its full evolutionary path through paleontology. Its history is instead inferred through comparative anatomy and genetic analysis of related species. (Werdelin et al 2010) Because the species was historically understudied, many aspects of its evolutionary history remain poorly documented, though recent studies are helping to fill in the gaps (Johnson et al., 2006). The Pallas's cat likely evolved as an early offshoot of small wildcats in Asia, adapting over millions of years to the cold, open landscapes of Central Asia. Its unique morphology and behaviors are the result of both environmental pressures and evolutionary isolation. A feature of the Pallas's cat that it evolved over time is its incredibly dense and fluffy fur, which is the thickest of any wild cat relative to its size. (Werdelin et al 2010) This plush coat is a critical adaptation to the harsh, cold environments of the Central Asian steppe. The fur provides exceptional insulation against freezing temperatures and biting winds, especially in high-elevation habitats that can reach up to 5,000 meters above sea level. Unlike many other small wildcats, the Pallas's cat has long guard hairs that create a rounded, puffed-up appearance, helping to retain body heat and shield it from the elements. Even its tail is bushy, offering additional warmth when curled around the body during rest. This signature fluffiness not only makes the Pallas's cat visually

distinctive but plays a vital role in its survival in one of the coldest, driest ecosystems on Earth (Greenspan & Giordano, 2021).

Section 3: Environmental Issues:

Some of the biggest factors affecting the survival of the Pallas's cat are human activities such as hunting and rodent extermination programs (Ross, 2009; Ross et al., 2010), food and habitat specialization (Ross et al., 2010), and fatal diseases such as toxoplasmosis (Brown et al., 2005).

Although the Pallas's cat is not currently classified as endangered, its conservation status frequently shifts between IUCN's Least Concern and Near Threatened due to habitat loss and declining population numbers (Ross et al., 2020). Among these threats, direct human impacts are particularly concerning. In Ross's 2009 study, he put GPS trackers on 30 individuals to observe their movements over a year.

Unfortunately over half of recorded Pallas's cat deaths were attributed to human causes such as poaching, hunting, automobiles, poisoning and domestic dogs. This level of anthropogenic mortality places serious pressure on the species especially because Pallas's cats already live in low population densities, making each individual loss more consequential (UN, 2024). Humans hunt Pallas's cats for their pelts, although hunting has been restricted by CITES international laws since 1988, illegal trade still happens. (Ross, 2009) In local Mongolian markets Pallas's cats fur is openly sold and parts of their bodies are occasionally used for traditional medicine. Thankfully their fur is not in high demand and doesn't sell for much. But hunting still caused 18% of the recorded deaths in the 2009 study. (Ross, 2009) It is estimated that the hunting-related cat deaths may be higher outside of Ross's study area.

Domestic dogs, however, were an even bigger problem, responsible for over 35% of all recorded deaths. Mongolian families typically own 2–3 dogs each, which they use to guard livestock, protect camps, and help hunt. The number of dogs increases in winter, sometimes reaching 18–28 dogs.

All of the dog-related deaths happened during these times of high dog density. Dogs are often only loosely supervised. They're rarely fed by people and often roam freely. They may be left with livestock or follow their owners around, and while wandering, they're more likely to come across and attack Pallas's cats. (Ross, 2009) Habitat degradation further compounds the threat. Human activities such as livestock grazing, agriculture, mining, and infrastructure expansion are increasingly encroaching on the Pallas's cat's native grassland and steppe environments (Ross et al., 2010). While some protected areas have been established in parts of Central Asia (Ross et al., 2020), the overall extent of undisturbed habitat continues to shrink. Because Pallas's cats prefer cold, dry, and sparsely vegetated areas, they are poorly adapted to survive in altered or fragmented landscapes (UN, 2024). This vulnerability is intensified by the cat's dietary specialization. Pallas's cats primarily prey on the Pallas's pika, a rodent that is often viewed as a pest in agricultural regions. To protect crops and livestock, Chinese and Mongolian farmers commonly poison these rodents. Unfortunately, Pallas's cats may consume the poisoned prey, leading to them becoming poisoned themselves and furthering population declines (Ross, 2009).

Moreover, the species faces ecological challenges due to being a den-dependent specialist. When rocky terrain is not available, the cat relies on shelters created by other animals in the grasslands, such as the endangered Siberian marmot (Pittet, 2024). As these burrow-building species decline due to habitat loss and climate change, Pallas's cats lose both shelter and hunting grounds (Ross et al., 2010). This tight ecological dependency creates a situation in which even small disruptions to the ecosystem can have cascading negative effects. Finally, since the Pallas's cat is a feline, it is susceptible to toxoplasmosis, a parasitic disease. Toxoplasmosis in most other cats is usually mild or asymptomatic, it can cause fever, lethargy, and loss of appetite in some cases. In kittens or immunocompromised cats,

the infection can be more severe, potentially affecting the lungs, liver, or nervous system. But in the Pallas's cat it is particularly susceptible. It is estimated that 60% of all Pallas's cat kittens die of the parasite and researchers don't have enough information on the cats to understand why yet. (Brown, 2005).

Because of the continuously dwindling population numbers, many feline-specific conservation organizations have been established to which help protect the Pallas's cat. These include groups such as the IUCN's *Species Survival Commission Cat Specialist Group* and the *Manul Working Group* (Pittet 2024) which specifically focuses on wild cats, as well as fan websites like *savemanul.org* or *manulization.com*. These conservation efforts are crucial, as the Pallas's cat faces threats from habitat degradation, climate change, and human encroachment (UN, 2024, Ross et. al., 2020). The IUCN's Cat Specialist Group conducts field research, develops conservation strategies, and works with local communities to reduce human-wildlife conflict (Ross et al., 2020). Meanwhile, fan-run platforms help raise awareness and build appreciation online for the species by sharing engaging content and supporting educational outreach. Both scientific expertise and public interest play a key role in ensuring the future of this elusive feline. In conclusion, survival of the Pallas's cat is increasingly threatened by a combination of human-driven impacts, ecological specialization, and habitat loss. While not yet officially endangered, their low population density, reliance on specific prey and habitats, and susceptibility to human caused mortality make them especially vulnerable. Conservation efforts must address both direct human threats such as hunting and habitat loss and the broader ecological challenges, including protecting prey species and preserving the integrity of their steppe habitats. By understanding the complexities between these factors, we can develop more targeted strategies to safeguard the future of this elusive and uniquely adapted wild cat.

Figure 1: “Bol” the Pallas’s cat in captivity at Nasu Animal Kingdom, Japan (Paulov, 2023)



Figure 2: Geographic Range. Primary residency in Mongolia. (Ross et al., 2020)

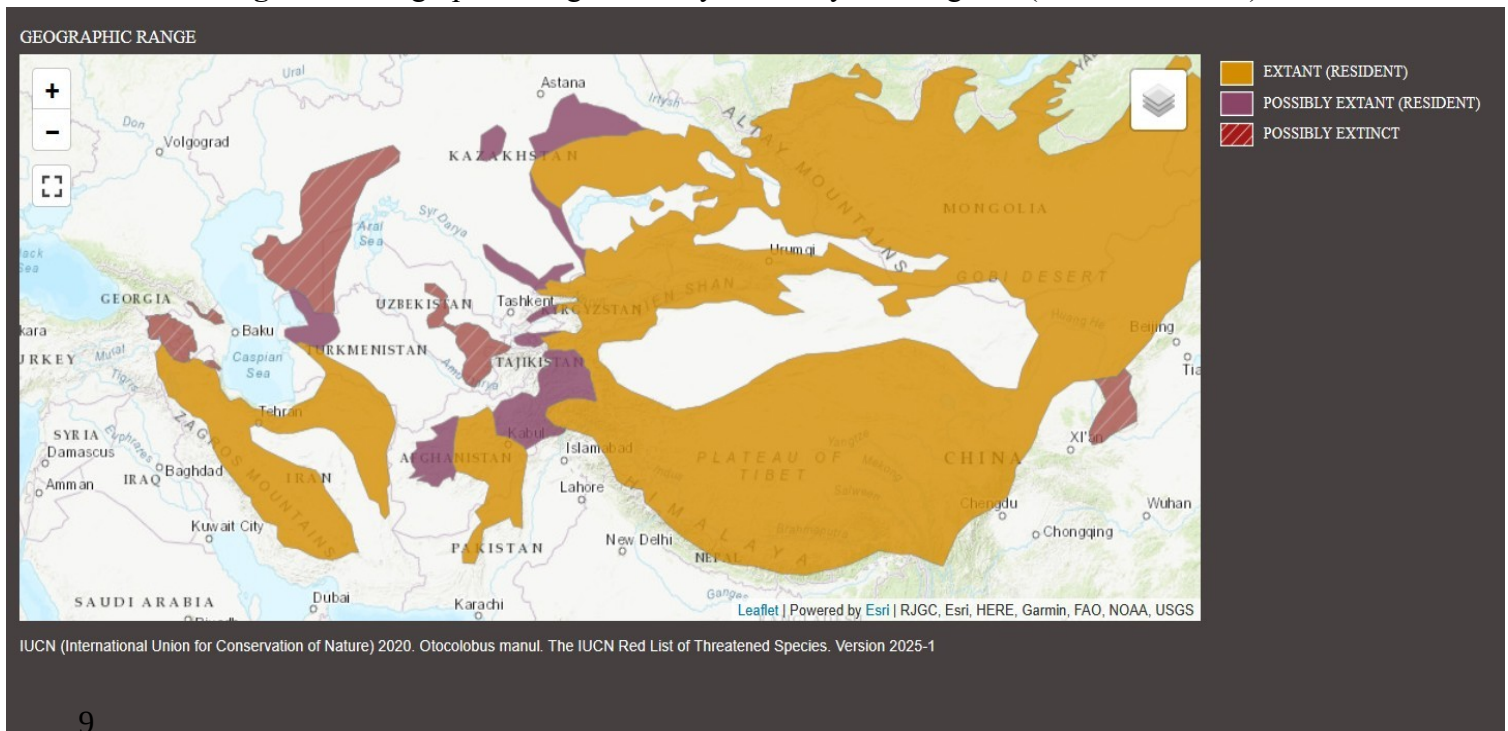
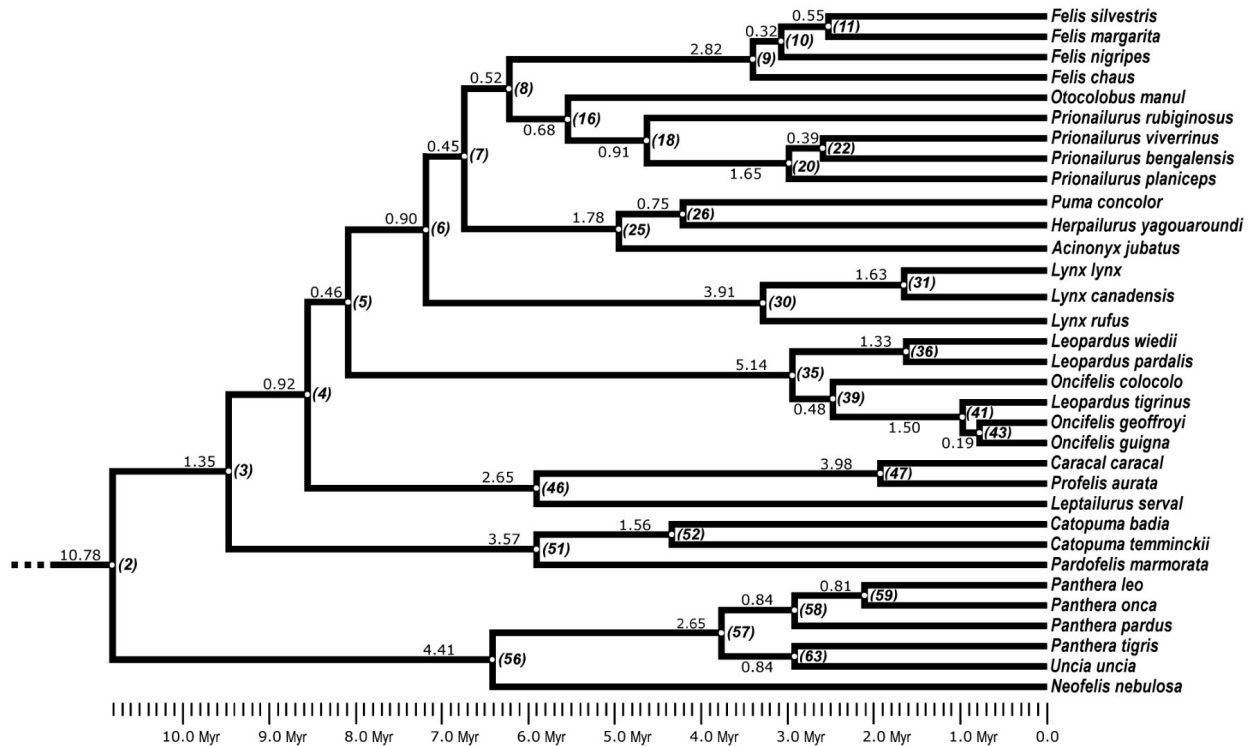


Figure 3: felidae phylogenetic tree containing *Otocolobus manul* (Sicuro, 2011)



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